

SAP S/4HANA ABAP Development

Exercise 21 – Adjust the User Interface

ABAP RESTful Application Programming Model (RAP)

Unit	8 – Describing the ABAP RESTful Application Programming Model
Exercise	21 – Adjust the User Interface
Objects	ZC_09FLIGHT – Behavior Projection / Metadata Extension
Status	COMPLETED – verified successfully

Objective

Explore the value help in the generated Fiori Elements application and adjust the UI through the RAP behavior projection and metadata extension.

Task 1 – Analyze the Value Help

- Open the OData UI service preview.
- Open a flight on the Object Page and switch to Edit mode.
- Open the value help for the Currency field.
- Inspect the projection CDS view ZC_09FLIGHT in Eclipse.

Annotation used:

```
@Consumption.valueHelpDefinition
```

Value-help CDS view:

```
I_CurrencyStdVH
```

Definition found in ZC_09FLIGHT:

```
@Consumption: {  
  valueHelpDefinition: [ {  
    EntityElement: 'Currency',  
    EntityName: 'I_CurrencyStdVH',  
    UseForValidation: true  
  } ]  
}  
CurrencyCode;
```

Task 2 – Adjust the Behavior

In the behavior projection ZC_09FLIGHT, Create and Delete were disabled for the UI service and PlaneTypeID was made read-only.

```
define behavior for ZC_09FLIGHT alias Flight  
{  
  field ( readonly ) PlaneTypeID;  
  
  // use create;  
  use update;
```

```
// use delete;

use action Edit;
use action Activate;
use action Discard;
use action Resume;
use action Prepare;
}
```

- Create disabled.
- Delete disabled.
- PlaneTypeID set to read-only.
- Behavior projection activated successfully.

Task 3 – Adjust the UI Metadata

The final metadata extension hides administrative fields, leaves only CarrierID and ConnectionID as selection fields, and places PlaneTypeID between FlightDate and Price.

```
@UI.identification: [ { position: 30 } ]
@UI.lineItem: [ { position: 30 } ]
FlightDate;
```

```
@UI.identification: [ { position: 40 } ]
@UI.lineItem: [ { position: 40 } ]
Price;
```

```
@UI.identification: [ { position: 35 } ]
@UI.lineItem: [ { position: 50 } ]
PlaneTypeID;
```

```
@UI.hidden: true
LocalCreatedBy;
```

```
@UI.hidden: true
LocalCreatedAt;
```

```
@UI.hidden: true
LocalLastChangedBy;
```

```
@UI.hidden: true
LocalLastChangedAt;
```

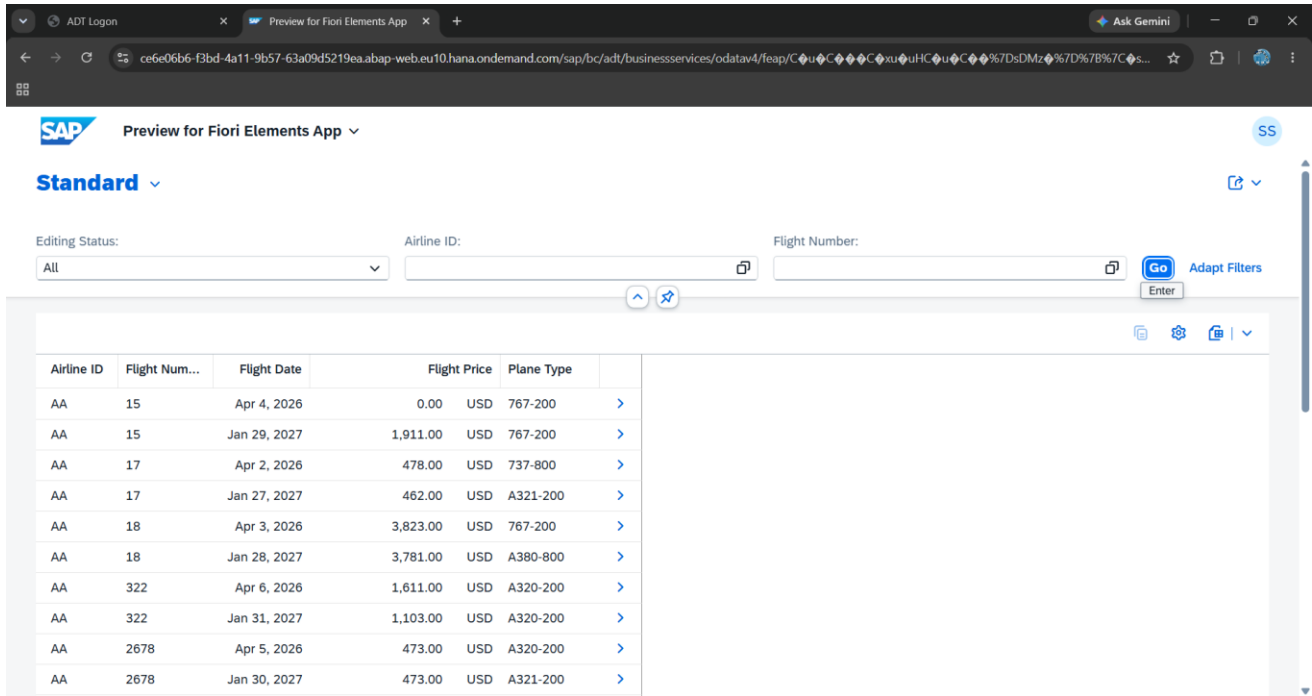
```
@UI.hidden: true
LastChangedAt;
```

- Only CarrierID and ConnectionID retain @UI.selectionField.
- Administrative fields use @UI.hidden: true.
- PlaneTypeID identification position is 35.
- Metadata extension activated with 0 errors.

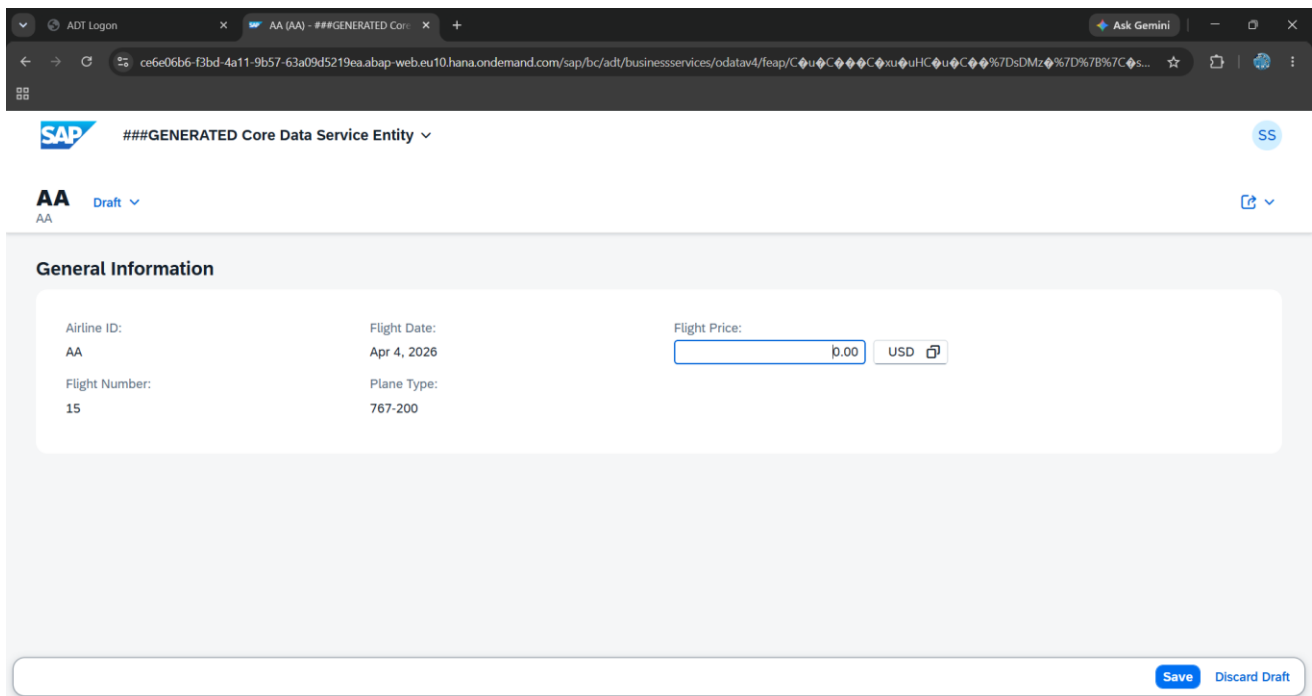
Task 4 – Verify the Result

The OData UI service preview was restarted and the final UI was checked.

Screenshot 1 – Report List Page



Screenshot 2 – Object Page in Edit Mode



Screenshot 1 confirms that Create/Delete are gone, administrative fields are hidden, and only Airline ID and Flight Number (the UI label for ConnectionID) remain as business selection fields.

Screenshot 2 confirms the Object Page order: Flight Date → Plane Type → Flight Price. Plane Type is read-only while Flight Price remains editable.

Final Verification Checklist

Requirement	Result
Create hidden	PASS

Delete hidden	PASS
Administrative fields hidden	PASS
Only CarrierID and ConnectionID are selection fields	PASS
PlaneTypeID between FlightDate and Price	PASS
PlaneTypeID read-only	PASS
Flight Price editable	PASS
Metadata activated with 0 errors	PASS

Key Concepts

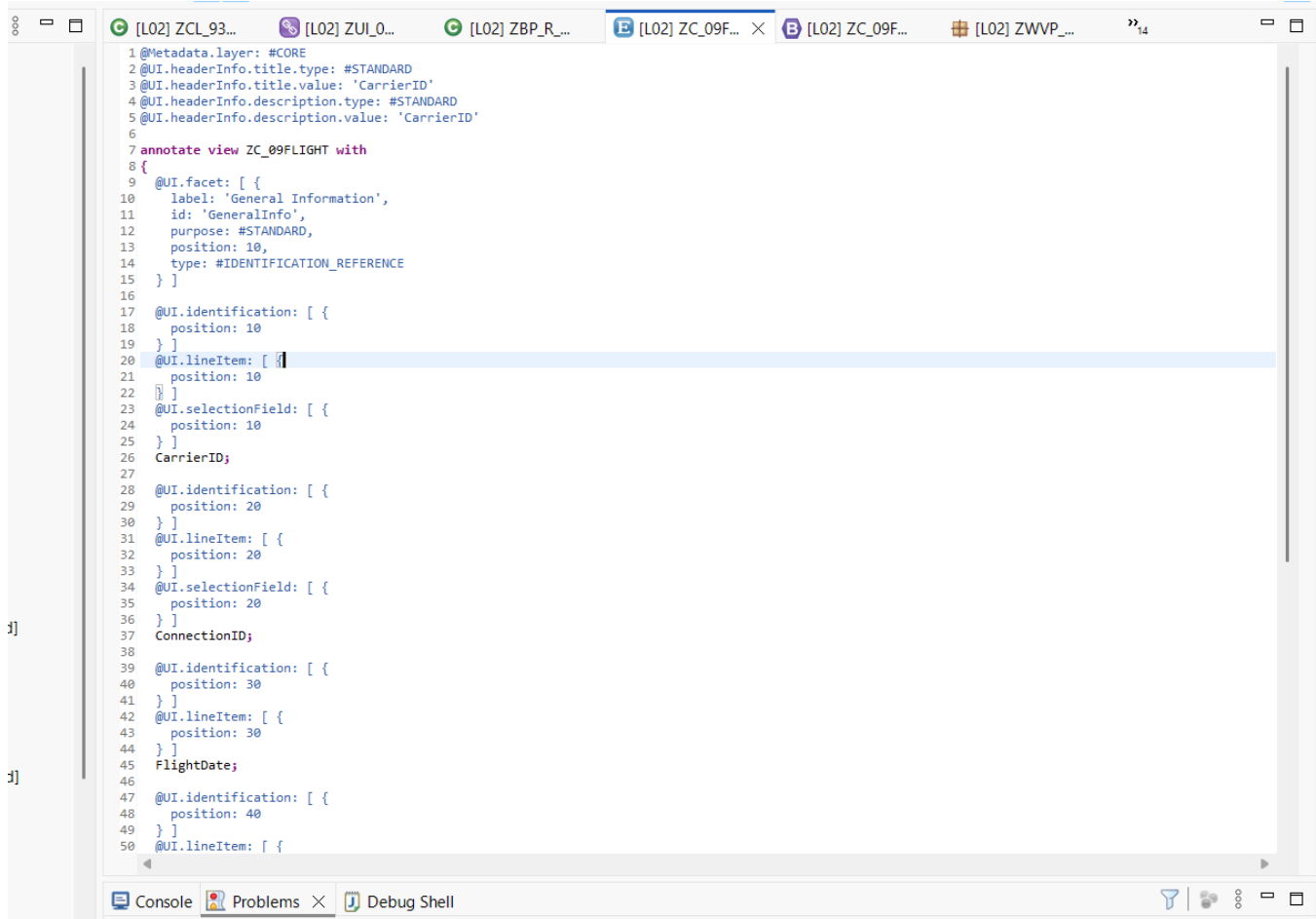
- @Consumption.valueHelpDefinition connects a field with a value-help CDS view.
- Behavior projections can restrict operations exposed by a particular service.
- field (readonly) makes a field non-editable in the exposed behavior.
- @UI.hidden controls field visibility.
- @UI.selectionField controls the list-page filter fields.
- @UI.identification controls Object Page field sequence.

EXERCISE 21 – COMPLETED

Exercise 21 – Eclipse Implementation Evidence

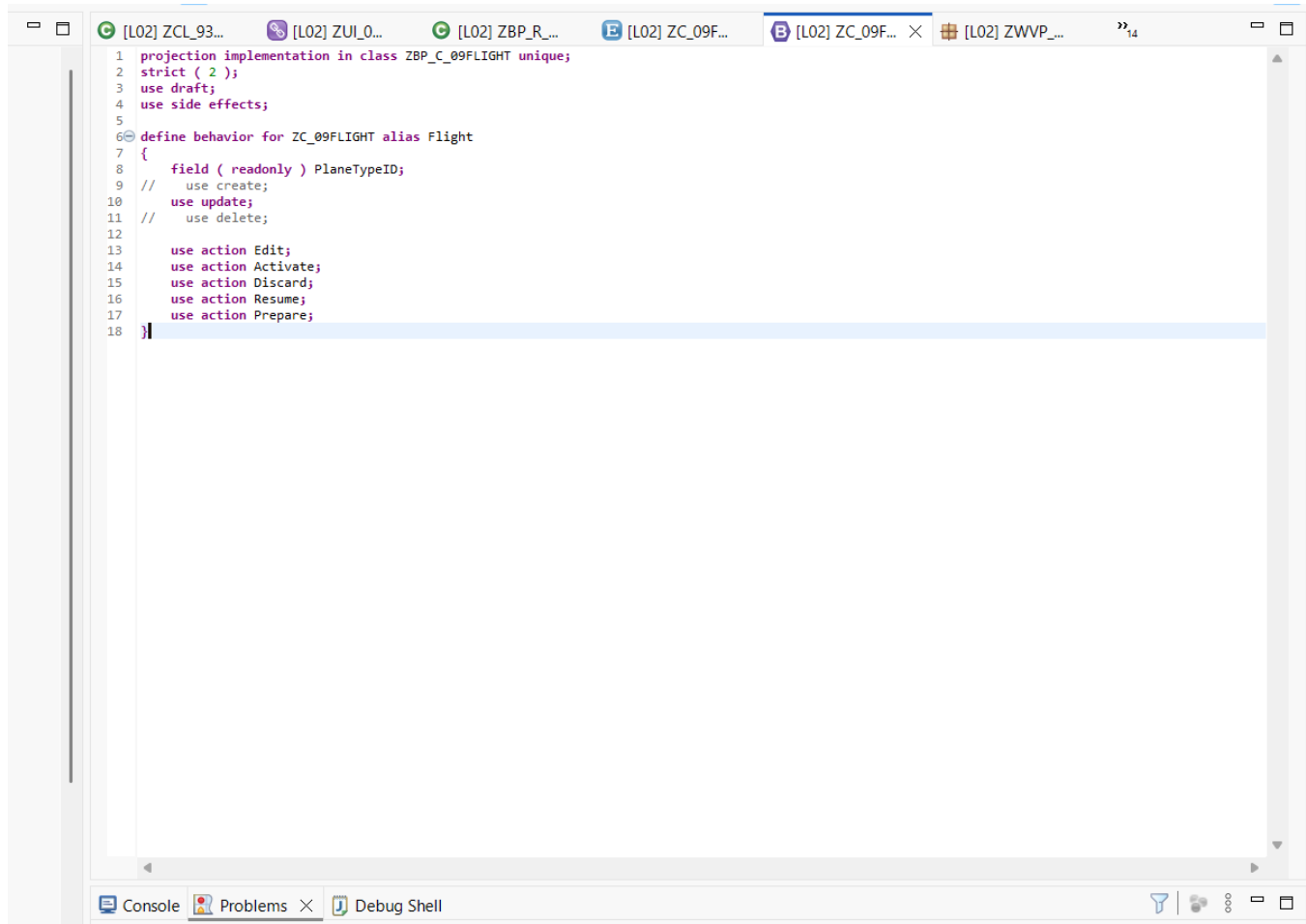
The following screenshots are directly relevant to Exercise 21. They show the final metadata extension and behavior projection used for the UI adjustments.

Screenshot 3 – Final Metadata Extension



```
1 @Metadata.layer: #CORE
2 @UI.headerInfo.title.type: #STANDARD
3 @UI.headerInfo.title.value: 'CarrierID'
4 @UI.headerInfo.description.type: #STANDARD
5 @UI.headerInfo.description.value: 'CarrierID'
6
7 annotate view ZC_09FLIGHT with
8 {
9   @UI.facet: [ {
10     label: 'General Information',
11     id: 'GeneralInfo',
12     purpose: #STANDARD,
13     position: 10,
14     type: #IDENTIFICATION_REFERENCE
15   } ]
16
17   @UI.identification: [ {
18     position: 10
19   } ]
20   @UI.lineItem: [ {
21     position: 10
22   } ]
23   @UI.selectionField: [ {
24     position: 10
25   } ]
26   CarrierID;
27
28   @UI.identification: [ {
29     position: 20
30   } ]
31   @UI.lineItem: [ {
32     position: 20
33   } ]
34   @UI.selectionField: [ {
35     position: 20
36   } ]
37   ConnectionID;
38
39   @UI.identification: [ {
40     position: 30
41   } ]
42   @UI.lineItem: [ {
43     position: 30
44   } ]
45   FlightDate;
46
47   @UI.identification: [ {
48     position: 40
49   } ]
50   @UI.lineItem: [ {
```

Screenshot 4 – Behavior Projection

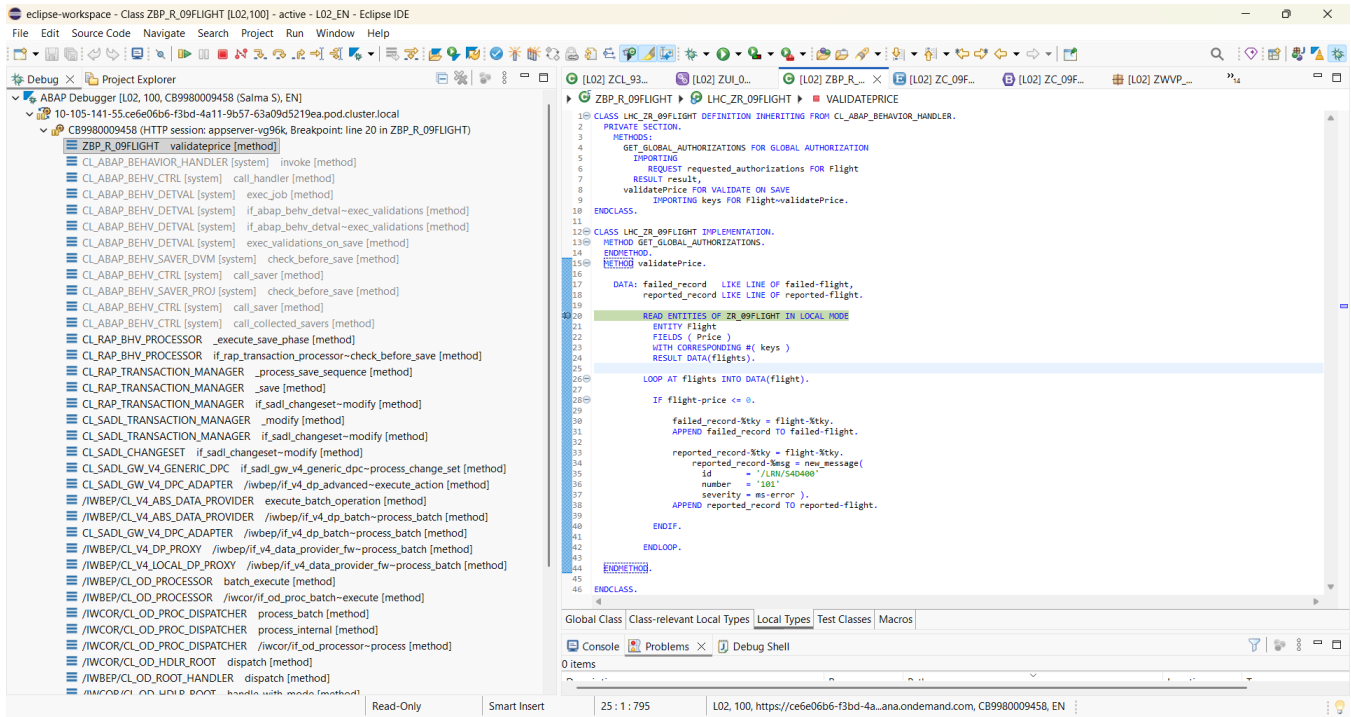


```
1 projection implementation in class ZBP_C_09FLIGHT unique;
2 strict ( 2 );
3 use draft;
4 use side effects;
5
6 define behavior for ZC_09FLIGHT alias Flight
7 {
8     field ( readonly ) PlaneTypeID;
9     // use create;
10    use update;
11    // use delete;
12
13    use action Edit;
14    use action Activate;
15    use action Discard;
16    use action Resume;
17    use action Prepare;
18 }
```

Metadata screenshot relevance: confirms CarrierID and ConnectionID retain @UI.selectionField, while FlightDate, PlaneTypeID and Price use the required @UI.identification positions. It also shows the Exercise 21 metadata extension.

Behavior screenshot relevance: confirms field (readonly) PlaneTypeID and that use create; and use delete; are commented out in the service behavior projection.

Related Existing Implementation – Validation Debugger



This debugger screenshot is related to the same Flight RAP business object and shows the existing validatePrice validation implementation. It is useful as project context, but it is not a change required by Exercise 21, so it is kept separate from the Exercise 21 task evidence.

Exercise 21 – Documentation Updated